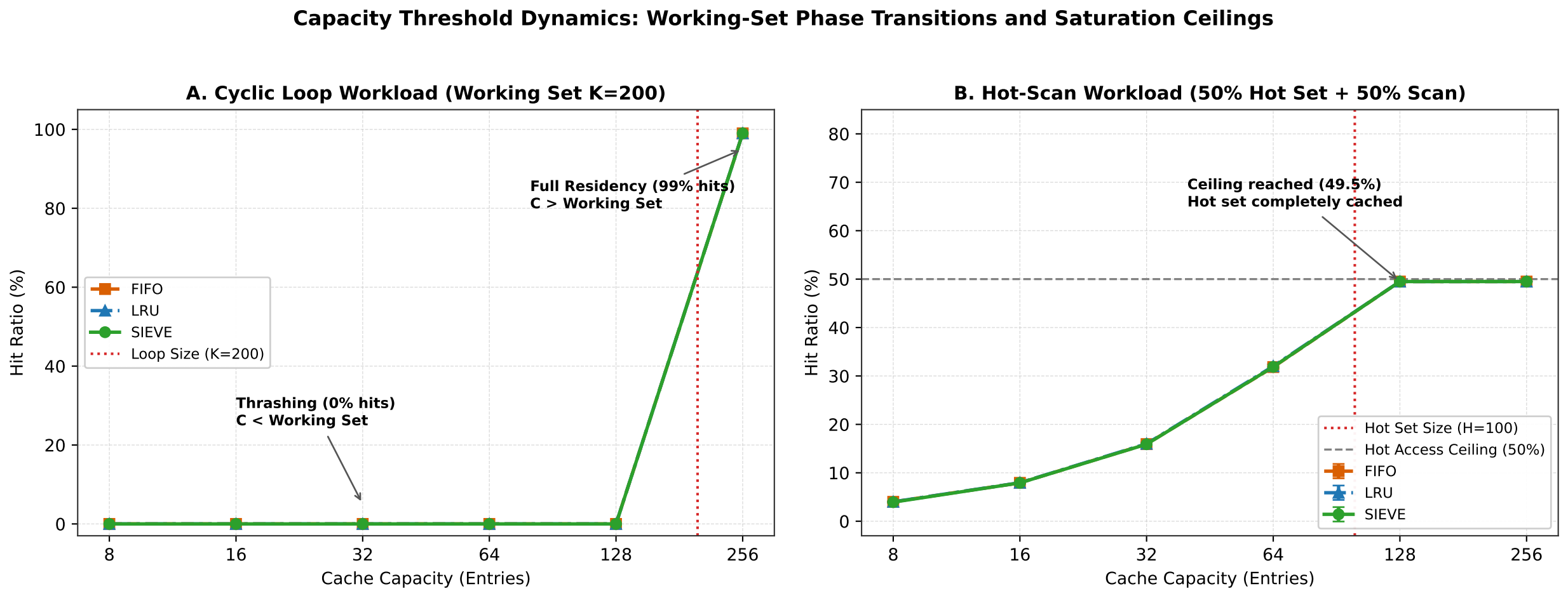
Capacity Threshold Dynamics: Working-Set Phase Transitions and Saturation Ceilings
A. Cyclic Loop Workload (Working Set K=200)
Hit Ratio (%)
Cache Capacity (Entries)
FIFO
LRU
SIEVE
Loop Size (K=200)
Thrashing (0% hits)
C < Working Set
Full Residency (99% hits)
C > Working Set
B. Hot-Scan Workload (50% Hot Set + 50% Scan)
Hit Ratio (%)
Cache Capacity (Entries)
Hot Set Size (H=100)
Hot Access Ceiling (50%)
FIFO
LRU
SIEVE
Ceiling reached (49.5%)
Hot set completely cached